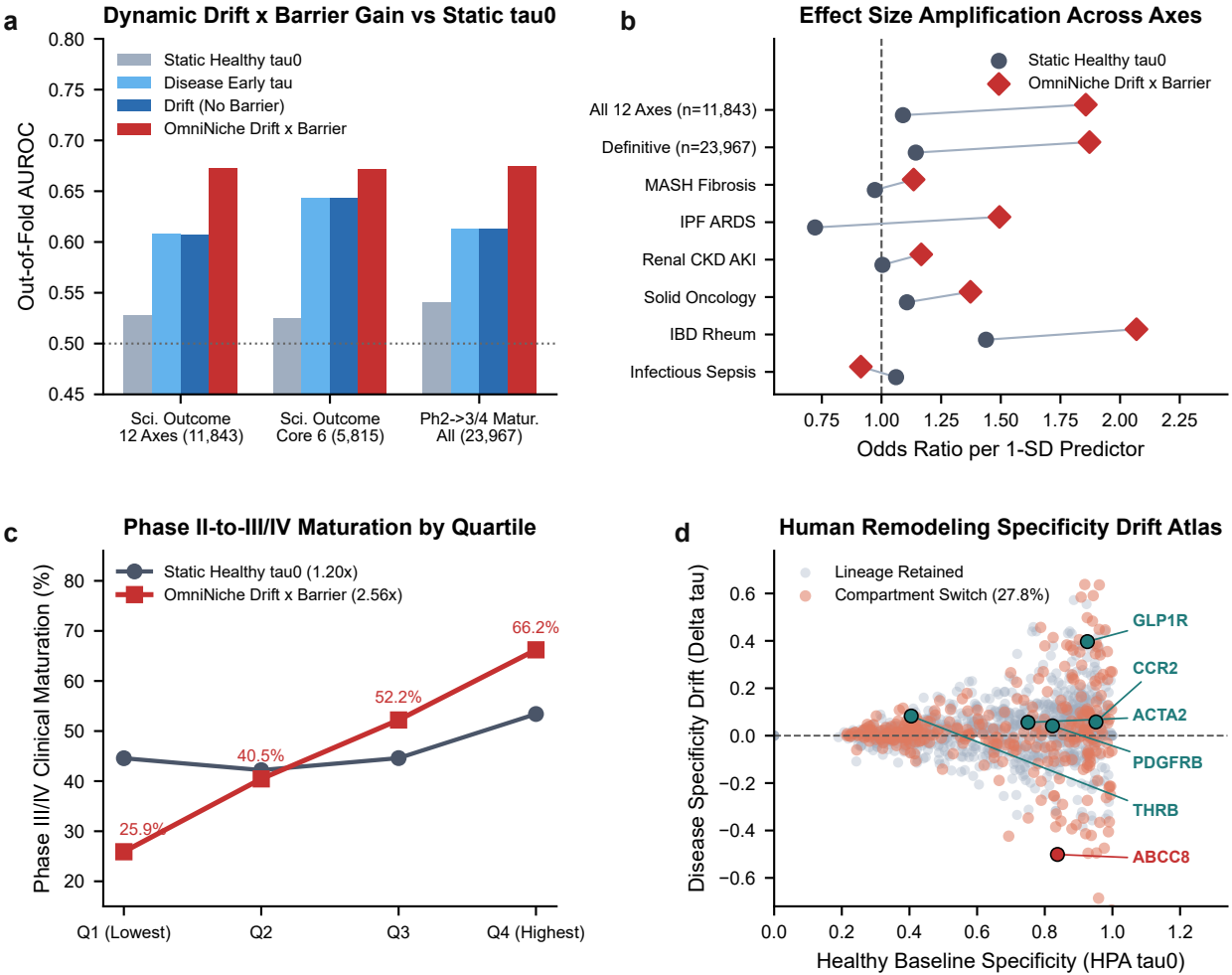

a
Dynamic Drift x Barrier Gain vs Static tau0
Static Healthy tau0
Disease Early tau
Drift (No Barrier)
OmniNiche Drift x Barrier
Out-of-Fold AUROC
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.45
Sci. Outcome 12 Axes (11,843)
Sci. Outcome Core 6 (5,815)
Ph2->3/4 Matur. All (23,967)
b
Effect Size Amplification Across Axes
Static Healthy tau0
OmniNiche Drift x Barrier
All 12 Axes (n=11,843)
Definitive (n=23,967)
MASH Fibrosis
IPF ARDS
Renal CKD AKI
Solid Oncology
IBD Rheum
Infectious Sepsis
0.75
1.00
1.25
1.50
1.75
2.00
2.25
Odds Ratio per 1-SD Predictor
c
Phase II-to-III/IV Maturation by Quartile
Static Healthy tau0 (1.20x)
OmniNiche Drift x Barrier (2.56x)
Phase III/IV Clinical Maturation (%)
80
70
60
50
40
30
20
66.2%
52.2%
40.5%
25.9%
Q1 (Lowest)
Q2
Q3
Q4 (Highest)
d
Human Remodeling Specificity Drift Atlas
Lineage Retained
Compartment Switch (27.8%)
GLP1R
CCR2
ACTA2
PDGFRB
THRB
ABCC8
Disease Specificity Drift (Delta tau)
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Healthy Baseline Specificity (HPA tau0)